

TUYIRINGIRE Pacifique (Paccy)

Computer Science Educator & Software Developer

"Transforming Learning Through Passion and Technology"

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PROFESSIONAL SUMMARY

Computer Science with Education student at the University of Rwanda with formal secondary school teaching experience, full-stack software development expertise, and a commitment to instructional technology. Experienced in Competency-Based Curriculum (CBC) delivery, resource-resilient pedagogy (instructing 68+ students and rotating 18 learners through 5–6 functional laptops), offline architecture engineering (C# / WPF, PWA), and multi-role institutional web/mobile platforms.

EDUCATION

Bachelor of Education – Computer Science with Education

(Current)

University of Rwanda – College of Education

Department of Mathematics & Computer Science Education

Key Coursework: Computer Science Pedagogy, Operating Systems, Algorithm Analysis & Design, Database Systems, Software Requirements (SRS), CBC Teaching Methodology, Educational Data Analytics.

TEACHING & PRACTICUM EXPERIENCE

Computer Science Teacher Intern

Jan 2025 – Present

GS MUHORORO — Murambi Sector, Rulindo District

- Facilitated senior secondary CS (S4 HGL in HTML/Web Development) and O-Level ICT (S3, 68 students including SEN learners).
- Pioneered resource-resilient pedagogy managing a Positivo BGH fleet of only 5–6 functioning student laptops for cohorts of 18–68 students.
- Employed Driver/Navigator pair coding, interactive projector code walkthroughs, and blackboard algorithm dry-runs to ensure 100% practical participation.
- Designed differentiated lesson plans aligned with Rwanda Basic Education Board (REB) Competency-Based Curriculum (CBC).

Teaching Assistant (Practicum Fellow)

2023 – 2024

Inspire, Educate and Empower Rwanda (IEE)

- Assisted secondary educators in classroom facilitation, remedial lesson design, and student-centered active learning methods.
- Supported community-school digital literacy programs and foundational tech integration in rural education centers.

SELECTED SOFTWARE SYSTEMS

HLI Timetable System [React, TypeScript, Constraint Satisfaction Solver]

Solves NP-hard scheduling bottlenecks at University of Rwanda, eliminating concurrent lecturer/hall overlaps across multiple student cohorts.

FluentEdge Academy & Certificate Ledger [React, Node.js, Cryptographic Verification Hash]

Comprehensive English learning hub featuring CEFR-aligned assessments, live progress telemetry, and tamper-proof certificate credentialing.

QuizMaster V2 (Offline LAN Assessment System) [C#, .NET, TCP Socket Server, SQLite]

Reliable offline computer-based examination system broadcasting questions over local Wi-Fi to 60+ concurrent workstations with automated gradebook CSV export.

Gasabo District Local Tax Collector [React Native, PostgreSQL, REST API]

Mobile fiscal tracking application designed for tax agents with offline caching, receipt generation, and transaction auditing.

ImbutoBooks Learning Reader [React, PWA, Service Workers, IndexedDB]

Ultra-low-bandwidth offline textbook and syllabus reader designed for students with limited internet connectivity.

TECHNICAL SKILLS & PEDAGOGICAL COMPETENCIES

Programming & Languages: C#, JavaScript (ES6+), TypeScript, Python, PHP, C, Java, SQL, HTML5, CSS3.

Frameworks & Mobile: React.js, React Native, Vite, Node.js, Express.js, WPF (.NET Desktop).

Databases & Tools: PostgreSQL, MySQL, SQLite, Git, GitHub, Linux Shell, Postman, Vite.

CS Pedagogy & Methodology: Competency-Based Curriculum (CBC), 5E Instructional Model, Driver/Navigator Pair Programming, Differentiated Learning, SEN Support.

LEADERSHIP & RECOGNITION

Mastercard Foundation Scholars Program (SEF 2.0 Fellow): Selected for leadership, academic excellence, and commitment to transformative education in Africa.

Huye Hackathon 2024 Semi-Finalist: Developed innovative social-impact tech solution addressing community bottlenecks under

incubation mentorship.

Choir President & Conductor: Christ the King Parish (Saint Paul Choir) — Directing 40+ vocalists, leading weekly rehearsals, logistics, and liturgical service.